

Lesson 1: Introduction to Computer Concepts & History

Early Computing & Key Pioneers

- **Abacus:** Derived from the Greek word *abax* ("tabular form"). Invented in ancient Babylon between 300 to 500 BC; recognized as the world's first counting machine/calculator. Used sliding counters and rods for arithmetic computations and is still used in education (e.g., Japan, China).
- **Cranmer Abacus:** Invented by Tim Cranmer for visually impaired users to perform addition, subtraction, multiplication, and division.
- **Pascaline (1642):** Invented by Blaise Pascal; the first mechanical calculator (gear-driven, performed addition and subtraction).
- **Jacquard Loom (1801):** Invented by Joseph Marie Jacquard; used punched cards to control weaving patterns.
- **Charles Babbage:** The "Father of Computing". Born December 26, 1791.
 - *Difference Engine:* Built to tabulate polynomial functions and eliminate calculation errors using finite differences.
 - *Analytical Engine:* First-ever design for a programmable digital mechanical general-purpose computer with output devices.
- **Augusta Ada Lovelace:** First computer programmer; wrote an algorithm for the Analytical Engine and foresaw computers manipulating symbols, music, and art.
- **German Z3 (1941):** Designed by Konrad Zuse; early programmable digital computer used during WWII for aircraft and missile design (3–10 calculations/sec).
- **Colossus (1944):** Designed by Tommy Flowers at Bletchley Park; used in WWII to decipher German High Command messages.

Generations of Computers

- **1st Gen:** Vacuum tubes, wired circuits, binary machine language, large/bulky, high cost. *Examples:* ENIAC (weighed 30 tons, occupied an entire room), EDVAC (introduced stored-program concept), UNIVAC (first commercial computer sold in the US).
- **2nd Gen (1956–1963):** Transistors; faster, smaller, less heat, more reliable, lower power draw. Introduced Assembly Language, FORTRAN, COBOL. *Examples:* IBM 7094, CDC 1604, IBM 1401.
- **3rd Gen (1964–1971):** Integrated Circuits (ICs) invented by Jack Kilby and Robert Noyce; monitors, keyboards, operating systems, and multiprogramming introduced. *Examples:* IBM System/360, PDP-8.
- **4th Gen (1971–Present):** Microprocessors, personal computers (PCs), GUIs, networking, large memory, laptops. *Examples:* IBM PC, Apple Macintosh.
- **5th Gen (Present & Beyond):** Artificial Intelligence (AI), Machine Learning, Robotics, NLP, Cloud Computing, Quantum Computing.

Types of Computers

- **Supercomputer:** Fastest and most powerful (e.g., Frontier).
- **Mainframe:** Used by large organizations requiring massive transaction processing.
- **Workstation:** High-performance PC designed for specialized professional or commercial use.
- **Desktop:** Stationary PC with separate base unit, monitor, keyboard, and mouse.
- **Laptop & Tablet:** Portable personal computers; laptops combine components and run on battery; tablets are book-sized touch devices.
- **Server:** Manages network resources, hosts websites, and serves client machines.

- **Smartphone & Gaming Console:** Handheld devices for communication/apps; consoles are dedicated devices with optimized gaming hardware.

Lesson 2: What is a Computer? Data & File Concepts

- **Definition:** An electronic device that takes raw data as input, processes it via programmed instructions, produces output, and stores results. Derived from *compute* ("to calculate").
- **Key Characteristics:** Speed, Accuracy, Storage Capacity, Automation, Diligence, Versatility, Reliability, Connectivity.
- **Basic Components:** Hardware, Software, Peopleware/User, Data, Procedures.
- **Five Functions:** Input, Processing, Output, Storage, Communication.
- **Data vs. Information:**
 - *Data:* Raw facts, symbols, or uninterpreted observations (e.g., numbers, names, dates).
 - *Information:* Processed, structured, and meaningful data ready for use.
 - *Data* *Processing* *Cycle:*
Data → Input → Processing → Information → Storage/Communication.
 - *Characteristics of Good Information:* Accuracy, Relevance, Timeliness, Completeness, Reliability, Understandability, Accessibility, Security.
- **Files & Folders:**
 - *File:* A collection of data stored as a single named unit with an icon and extension.
 - *Folder:* A directory container used to group and organize files and subfolders.
 - *Extensions:* 3–4 letter tags indicating file type: **.docx/.doc** (Word), **.xlsx/.xls** (Excel), **.pptx/.ppt** (PowerPoint), **.jpg/.jpeg** (images), **.mp3** (audio), **.html** (web), **.exe** (executables), **.pdf**, **.zip**, **.csv**, **.psd**.

Lesson 3: Computer Hardware

- **Definition:** The tangible, physical parts of a computer system.
- **Input Devices:** Provide data and control signals to the computer (e.g., Keyboard, Mouse, Scanner, Microphone, Webcam).
- **Output Devices:** Convey processed information to users:
 - *Printers:* Dot Matrix, Inkjet, Laser Jet, 3D Printer.
 - *Monitors:* CRT (Cathode Ray Tube), LCD (Liquid Crystal Display), LED (Light Emitting Diode), OLED (Organic LED).
 - *Others:* Projectors, Speakers.
- **Internal Components & System Unit:**
 - *System Unit / Chassis:* The protective case housing motherboard, CPU, RAM, and internal devices.
 - *Motherboard:* The main circuit board that connects all computer components together.
 - *CPU:* The "brain" that executes program instructions.
 - *Cooling:* Heat sink absorbs heat, fan dissipates it; thermal paste fills microscopic air gaps between CPU and heatsink.
 - *Storage:*
 - *HDD:* Mechanical drive with platters, spindle, and actuator arms.
 - *SSD (Solid State Drive):* Faster, shock-resistant storage using NAND flash memory chips and a controller.
 - *Cables:* SATA (Serial ATA) and legacy PATA/IDE cables.

- *Memory*:
 - *RAM (Random Access Memory)*: Volatile, temporary working memory.
 - *ROM (Read Only Memory)*: Non-volatile, permanent storage chip.
- *Expansion Cards & Power*: Video/Graphics Card (renders images to displays), Optical Drive (uses lasers for discs), Power Supply Unit (PSU - delivers electricity).

Lesson 4: Computer Software

- **Definition**: A collection of instructions, data, or programs that direct computer hardware to execute tasks.
- **Systems Software**:
 - Controls hardware operations and creates the foundation for application software.
 - Acts as the intermediary/interface between the user and physical hardware.
 - *Operating System (OS)*: The principal software coordinating all components (e.g., Windows, macOS, Linux).
- **Application Software**:
 - Programs designed to assist users in completing personal, educational, or business tasks.
 - *Web Browsers*: Chrome, Firefox, Safari, Edge, Brave.
 - *Word Processing*: MS Word, Google Docs, OpenOffice Writer.
 - *Spreadsheets*: MS Excel, Google Sheets, OpenOffice Calc.
 - *Graphics*: Adobe Photoshop, GIMP, MS Paint.
 - *Presentations*: MS PowerPoint, Google Slides, Canva, OpenOffice Impress.
 - *Education & Collaboration*: LMS (Google Classroom, Edmodo), video conferencing (Zoom, Google Meet, Webex).